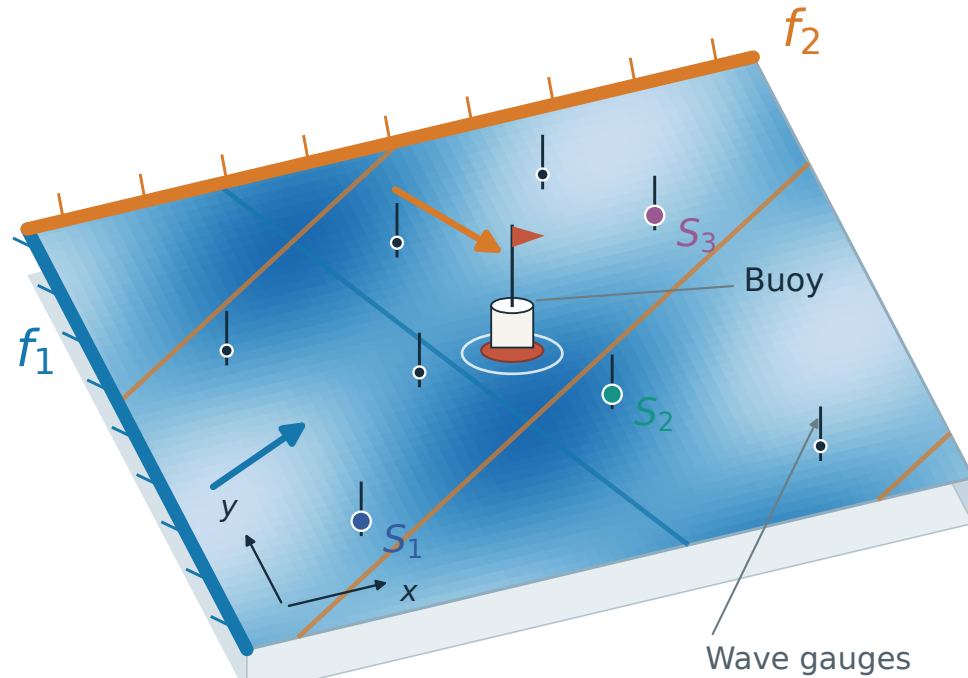


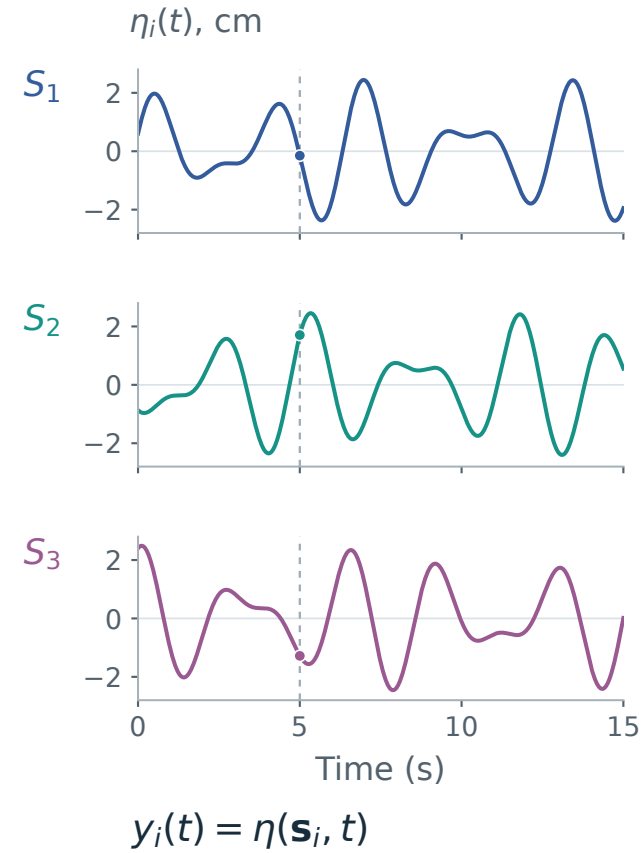
Crossing waves: from measurements to spatial phase

a Physical system

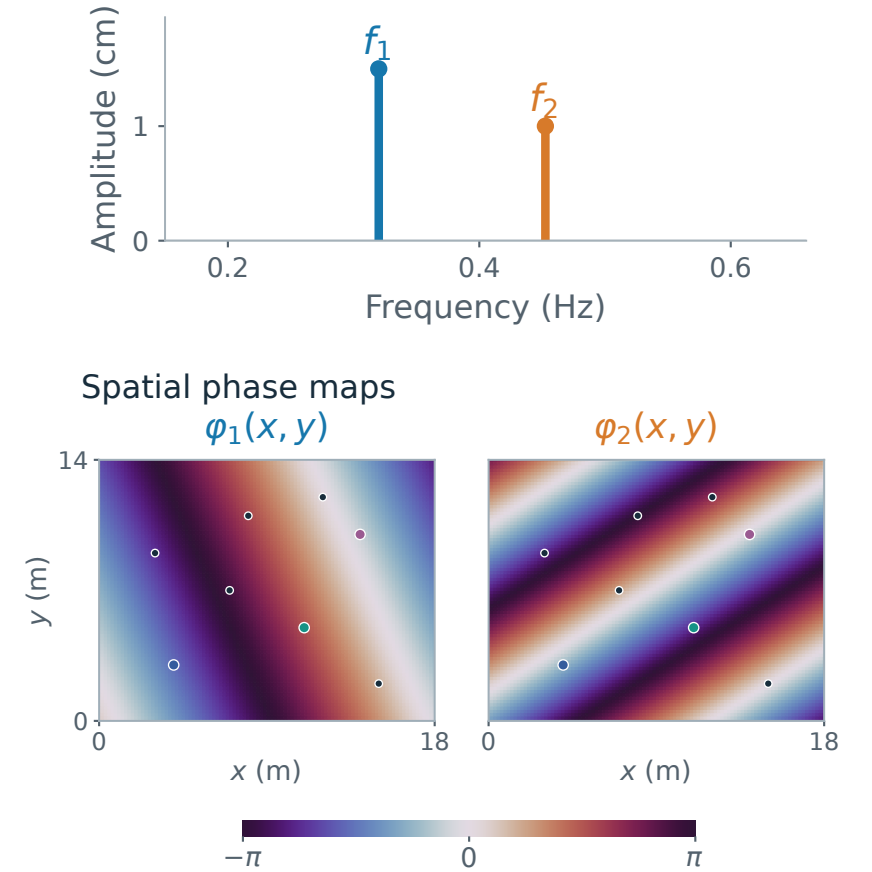
Two harmonic inputs



b Measurements



c Spectrum and phase



$$\eta(\mathbf{s}, t) = a_1 \cos[2\pi f_1 t + \phi_1(\mathbf{s})] + a_2 \cos[2\pi f_2 t + \phi_2(\mathbf{s})]$$

Illustrative model: small waves, deep water, no reflections. Gauge signals and phase maps are calculated from the same field.

$f_1 = 0.320$ Hz · $f_2 = 0.320\sqrt{2}$ Hz · $a_1 = 1.5$ cm · $a_2 = 1.0$ cm